# Welcome to the Workshop!

- 2 Day Series
  - Applying AI to **real business workflows**
- Today we'll cover some theory
  - We'll also **try some prompting strategies**

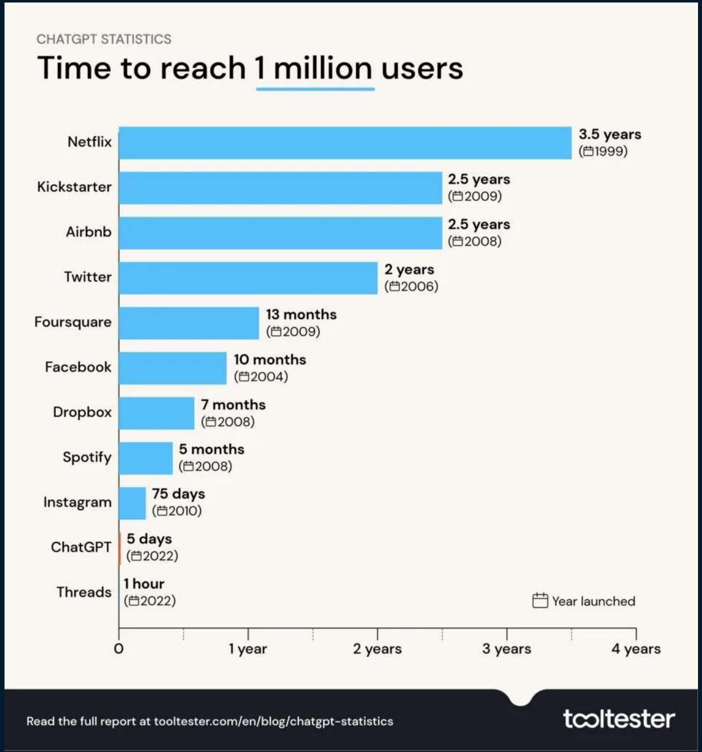
- **No code** required, just structured thinking
  - (and a little bit of **trial** and **error**)
- On Wednesday, you'll get in your project teams and **build your own AI-powered system**

01

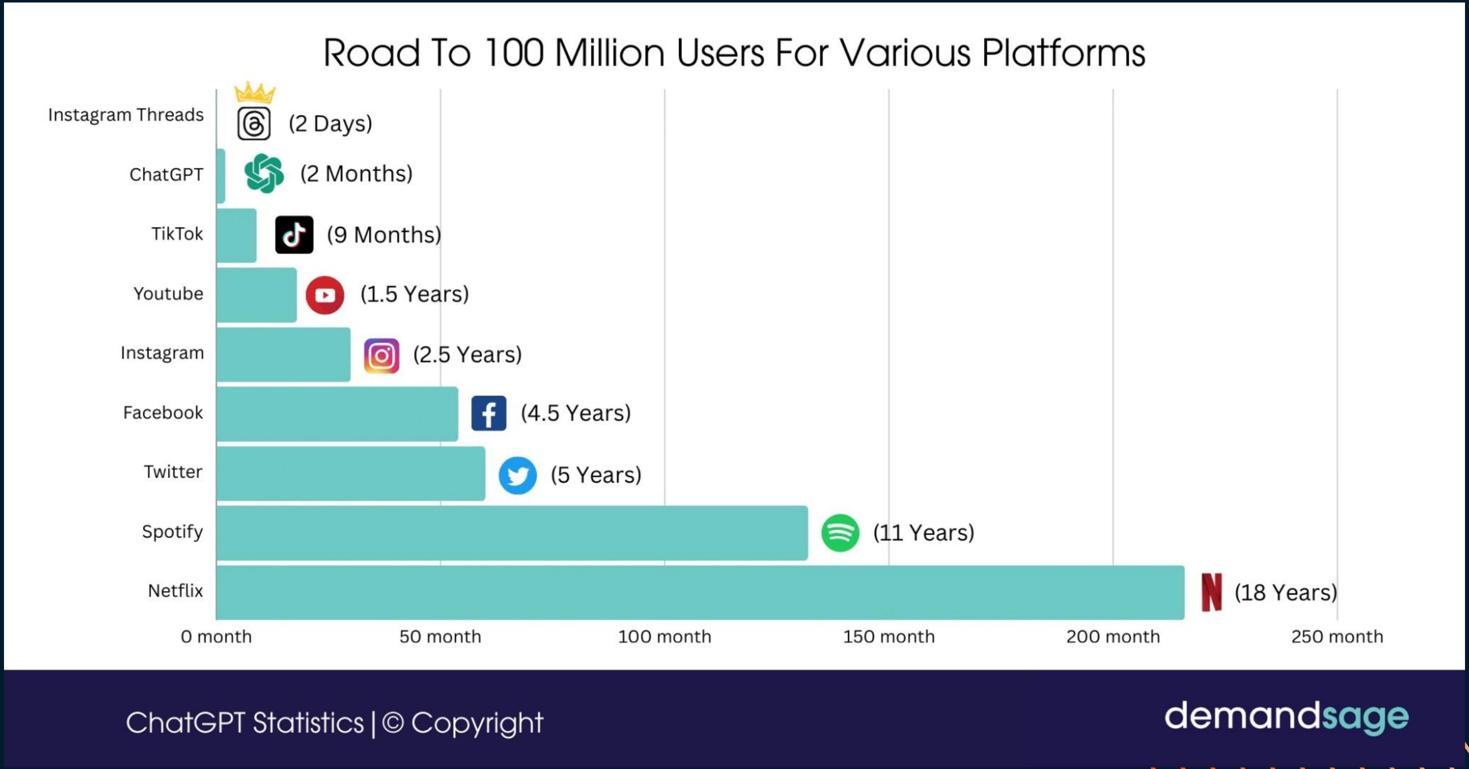
# Setting the Stage

What will we be covering, and why now?

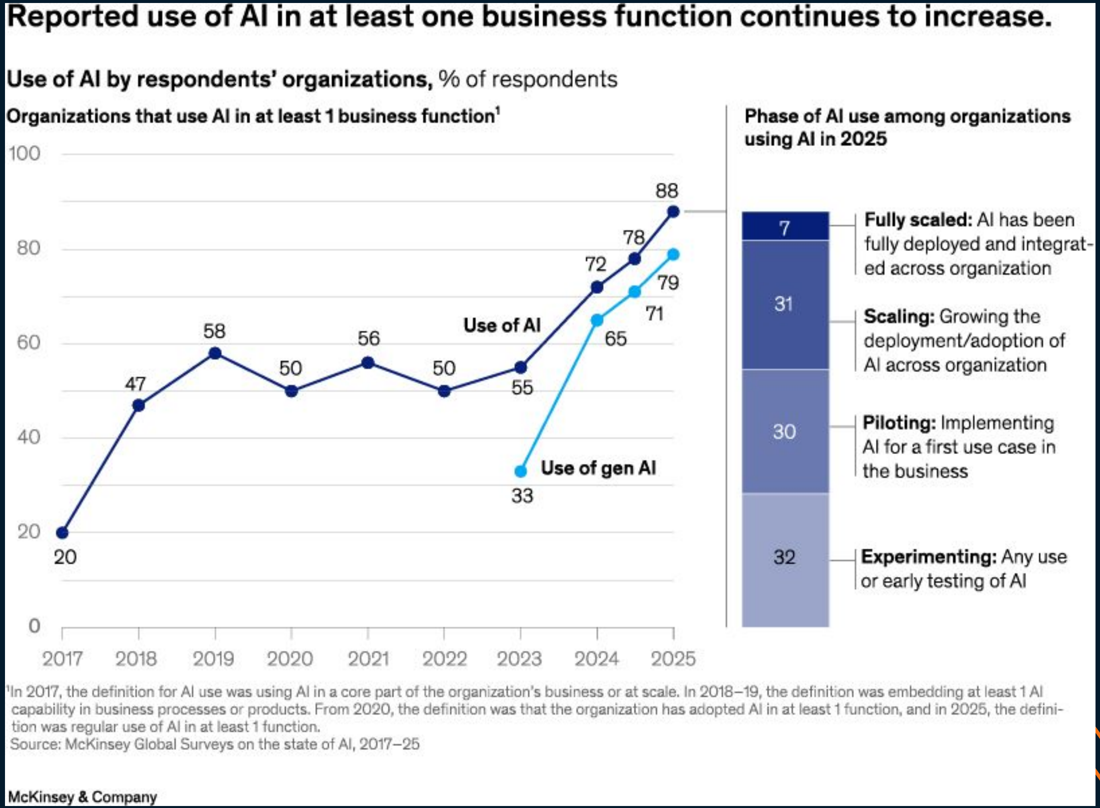
# Why Now?


CHATGPT STATISTICS
Time to reach 1 million users
Netflix
3.5 years
(1999)
Kickstarter
2.5 years
(2009)
Airbnb
2.5 years
(2008)
Twitter
2 years
(2006)
Foursquare
13 months
(2009)
Facebook
10 months
(2004)
Dropbox
7 months
(2008)
Spotify
5 months
(2008)
Instagram
75 days
(2010)
ChatGPT
5 days
(2022)
Threads
1 hour
(2022)
Year launched
0
1 year
2 years
3 years
4 years
Read the full report at tooltester.com/en/blog/chatgpt-statistics
tooltester

# Why Now?


Road To 100 Million Users For Various Platforms
Instagram Threads
(2 Days)
ChatGPT
(2 Months)
TikTok
(9 Months)
Youtube
(1.5 Years)
Instagram
(2.5 Years)
Facebook
(4.5 Years)
Twitter
(5 Years)
Spotify
(11 Years)
Netflix
(18 Years)
0 month
50 month
100 month
150 month
200 month
250 month
ChatGPT Statistics | © Copyright
demandsage

# Why Now?



Reported use of AI in at least one business function continues to increase.
Use of AI by respondents' organizations, % of respondents
Organizations that use AI in at least 1 business function[1]
100
80
60
40
20
0
2017
2018
2019
2020
2021
2022
2023
2024
2025
20
47
58
50
56
50
55
72
78
88
Use of AI
Use of gen AI
33
65
71
79
Phase of AI use among organizations using AI in 2025
7
31
30
32
Fully scaled: AI has been fully deployed and integrated across organization
Scaling: Growing the deployment/adoption of AI across organization
Piloting: Implementing AI for a first use case in the business
Experimenting: Any use or early testing of AI
[1]In 2017, the definition for AI use was using AI in a core part of the organization's business or at scale. In 2018–19, the definition was embedding at least 1 AI capability in business processes or products. From 2020, the definition was that the organization has adopted AI in at least 1 function, and in 2025, the definition was regular use of AI in at least 1 function.
Source: McKinsey Global Surveys on the state of AI, 2017–25
McKinsey & Company

# AI as Infrastructure

## Customer Support

Chatbots & Automatic Routing

## Product Planning

Review Summarizers, Customer Insights Analysis

## Sales Ops

Leads Scoring, Outreach Automation

## Onboarding & HR

Policy Assistant + Training Guides

# What You'll Be Doing

1. Pick a **role**
2. Choose a **knowledge source**
3. **Write + Refine** prompts to simulate agentic behavior
4. **Observe** outputs and **iterate**
5. Build a system with clear inputs and outputs

02

# A Brief Timeline

# 2015-2021 - The Slow Burn

*Nov 2015* **- Tensorflow** open sourced by Google

*Jun 2017* **- "Attention is All You Need"** published

*Jun 2018* **- GPT 1** introduced

*Oct 2018* **- BERT** introduced

*May 2020* **- GPT 3** introduced

# 2022 - Gen AI Emerges

*Jan 27* - **InstructGPT** released by OpenAI

*Jan 28* - **Chain-of-Thought** paper published

*Jun 11* - **Blake Lemoine** claims Google's LaMDA is sentient

*Oct 7* - US introduces **export controls on semiconductors** to China

*Nov 30* - **ChatGPT** released

# 2023 - Development Race

*Feb 24* - **LLaMA** released by Meta

*Mar 14* - **GPT 4** released

*Mar 14* - **Claude** released by Anthropic

*Apr 16* - **AutoGPT** open sourced on GitHub

*May 30* - **Nvidia** reaches a valuation of **$1 Trillion**

*Nov 1* - **UK AI Summit**

*Nov 17* - **Sam Altman** fired, and reinstated in **4 Days**

# 2024 and Beyond - AI Becomes Ubiquitous

*Mar 2024* - **Devin** released by Cognition Labs

*May 2024* - **GPT 4o** released

*May 2024* - **EU AI Act** enacted

*Oct 2024* - **Claude 3** released by Anthropic

*Jan 2025* - **Deepseek R1** open sourced

*Jan 2025* - **Stargate Project** announced

*2025 Onwards* - Continuing Development and Adoption

# 03

# From Gen AI to Agentic AI

# What is Agentic AI?

- Agentic AI describes AI systems that **autonomously** make decisions and **take actions**
- Agents are **proactive**, where GenAI is **reactive**
    - Ex: Getting to the Airport


Input
Tool Use
Output
Send Check-In Reminder
Calendar Event (Southwest Flight - SFO > JFK)
Check Traffic to Airport
"I have a flight at 8 AM to NYC"
15 Minute Delay Likely
Prebook Uber to Airport
Send Booking Confirmation
Push Notification: Uber Arriving Soon

# Key Difference?

Gen AI can give me a smart answer

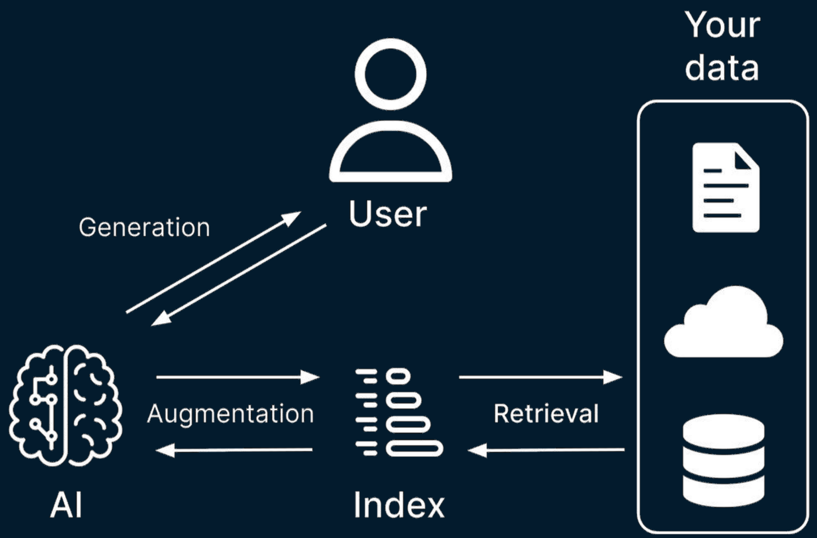
Agentic AI **took charge**, and got me to the airport

# 04

# Fundamentals of Grounding & RAG

# What is RAG?

- Adding external knowledge to **ground** an AI model's responses
- Instead of relying only on what the model "knows", we can give it:
    - Documents
    - Data
    - Or any other inputs
- Helps improve **accuracy, retention,** and **task specialization**

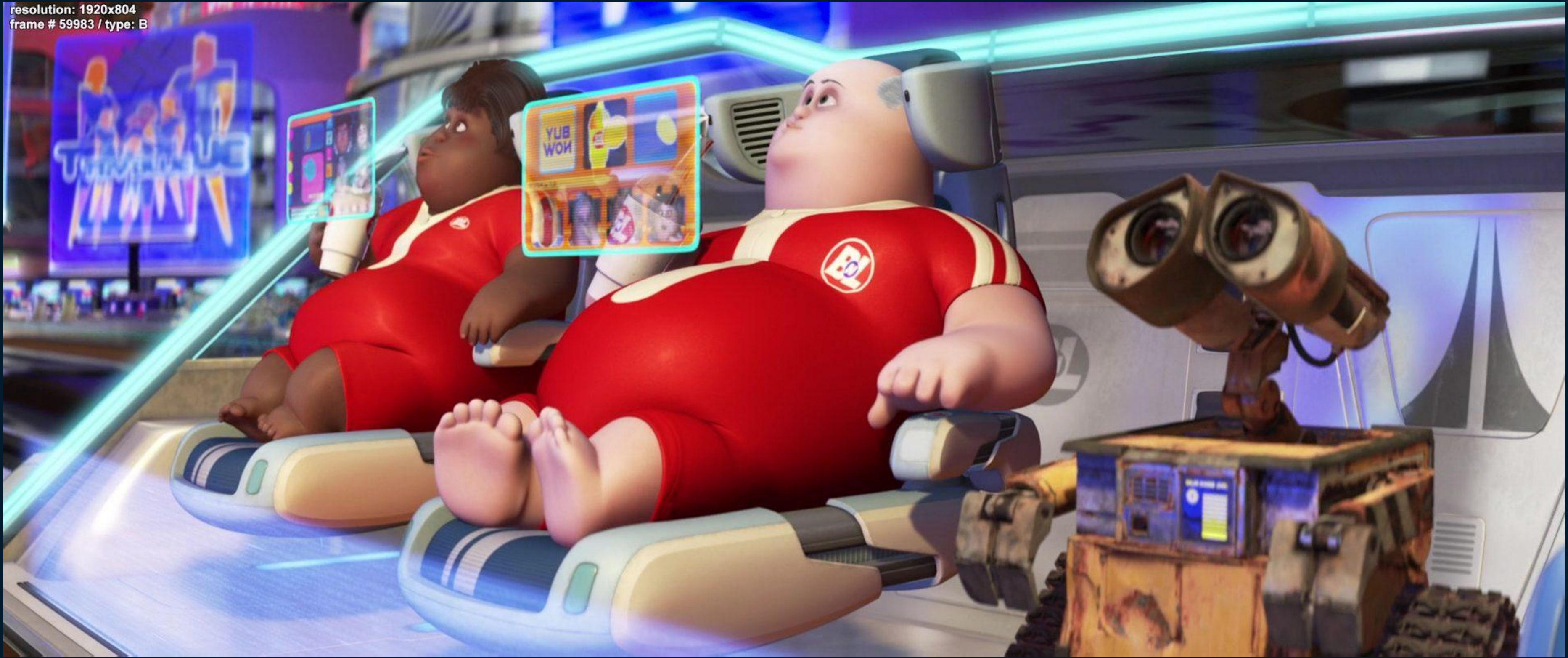


# Why RAG?

- LLMs (Large Language Models) are trained on **huge, general purpose** datasets
  - They're not experts in any one domain
  - And they'll guess if unsure
- We can use RAG to:
  - Ground answers in real, topical information
  - Support specialized tasks
  - Keep outputs accurate and current

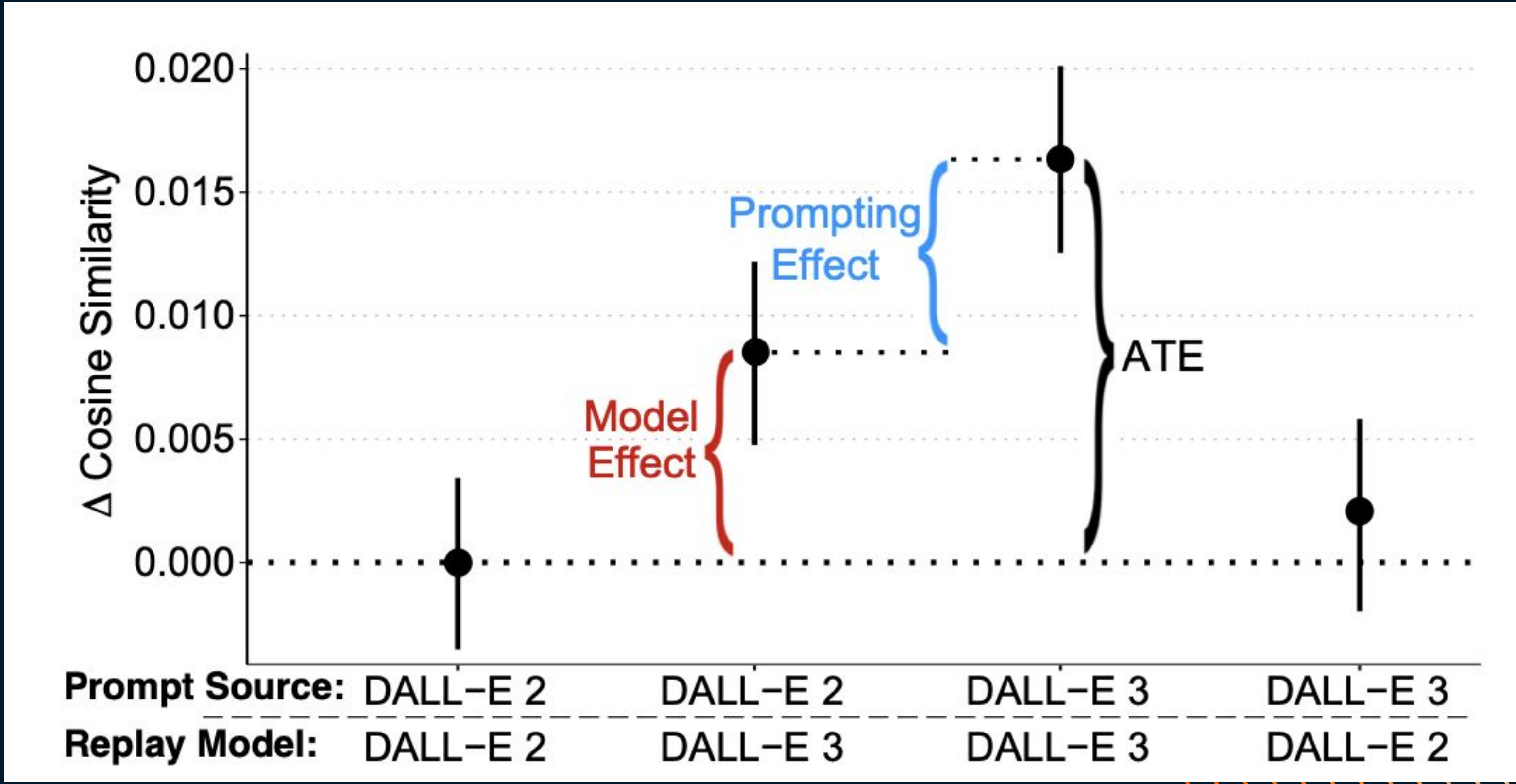
# Risks of Hallucinations

- Hallucinations are having real-world costs:
  - Med-Gemini made up **parts of the brain**
  - Publishing a summer reading list with **10 of 15** books made up
  - Chatbot makes up policy, **Air Canada held liable**
  - Widespread **fabrication of case law** in legal briefs
    - **1081** and counting!


resolution: 1920x804
frame # 59983 / type: B
BUY NOW

# Simulating Agentic AI with Prompt Engineering

# Why Does It Matter?



# What Makes a Good Prompt?

- Task - What do you want done?
- Context - What should the model know to do the task?
- Example - What does a good response look like?
- Persona - Who should the model act like/as?
- Format - How should the response be structured?
- Tone - How should the response sound?

# What Makes a Good Prompt?

| Element | Description & Tips |
|---|---|
| **Task** | Start with an **action verb** *(solve, write, analyze…)* |
| **Context** | Supply the background or environment |
| **Example** | Illustrate what "good" looks like |
| **Persona** | Tell the model who it's acting as |
| **Format** | Specify structure & constraints |
| **Tone** | Set voice or style |

# Anatomy of a Prompt

- Problem: Meal planning
  - Just asking “weekly foods” or “meal plan” won’t give you what you want
- Task:
  - Give me a 1 week plan for meals
- Context + Task:
  - I am a 180 lb male looking to maintain my weight. Give me a 1 week plan for meals.
- Context + Example + Task:
  - I am a 180 lb male looking to maintain my weight. Here’s an example of what one day should contain: Breakfast ... Lunch ... Snack ... Dinner. Give me a one week plan for meals

# Anatomy of a Prompt

- Context + Example + Task + Format:
  - I am a 180 lb male looking to maintain my weight. Here's an example of what one day should contain: Breakfast ... Lunch ... Snack ... Dinner. Give me a one week plan for meals in bullet points
- Persona + Context + Example + Task + Format:
  - Act as a Nutritionist and answer the following. I am a 180 lb male looking to maintain my weight. Here's an example of what one day should contain: Breakfast ... Lunch ... Snack ... Dinner. Give me a one week plan for meals in bullet points
- Tone + Persona + Context + Example + Task + Format:
  - In a professional tone: Act as a Nutritionist and answer the following. I am a 180 lb male looking to maintain my weight. Here's an example of what one day should contain: Breakfast ... Lunch ... Snack ... Dinner. Give me a one week plan for meals in bullet points

# How to Keep Your Prompts Grounded

- Use a grounding document or dataset
- Explicitly limit the model's knowledge scope
- Ask for quotes or source references

# Live Demo

Building a Customer Experience Analysis agent

# The Concept

- We'll simulate a Customer Experience Analysis agent
- Grounding Document:
  - Extracted Customer Reviews
- Tool:
  - ChatGPT (Web UI)
- Objective:
  - Extract top complaints
  - Reformat these insights for stakeholders

# Layering Prompt Elements

- Iteration 1: Task
- Iteration 2: Add Persona and Format
- Iteration 3: Adding Tone and Context

# Iteration 1 - Task

Summarize the top three most common complaints from the AirPods Pro 3 review digest.

# Iteration 2 - Persona + Task + Format

You are a customer experience analyst. Based only on the uploaded AirPods Pro 3 review digest, return three bullet points summarizing the most common customer complaints. Include one short supporting quote under each bullet.

# Iteration 3 - Persona + Context + Task + Format + Tone

You are a customer experience analyst preparing a summary for the product team. Using only the uploaded AirPods Pro 3 review digest, write a one-paragraph internal memo summarizing the top three user complaints. Keep the tone neutral, professional, and focused on actionable insights for the product team.

# Agents vs Workflows

# Two Layers of Intelligence

**Reasoning Layer - Agents**

Goal-driven systems that plan and take actions

**Execution Layer - Workflows**

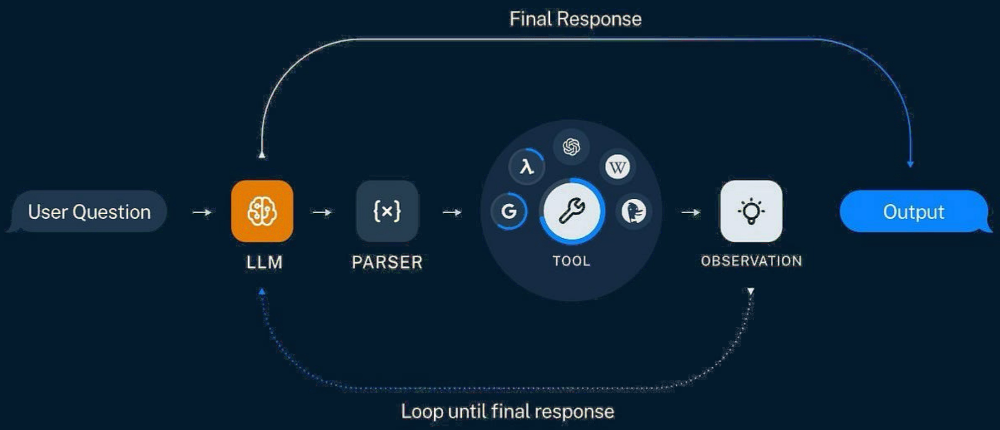
Defined, rule-based processes for repetition and consistency

# Key Differences

| Aspect | AI Workflows | AI Agents |
| --- | --- | --- |
| Control Flow | Predefined, explicit | Dynamic, autonomous |
| Decision Making | Hard-coded logic | LLM-driven reasoning |
| Tool Usage | Linear, orchestrated | Self-selected by agent |
| Adaptability | N/A; Fixed path | Flexible execution |
| Complexity | Lower, more predictable | Higher, more capable |
| Use Cases | Well-defined tasks | Open-ended problems |

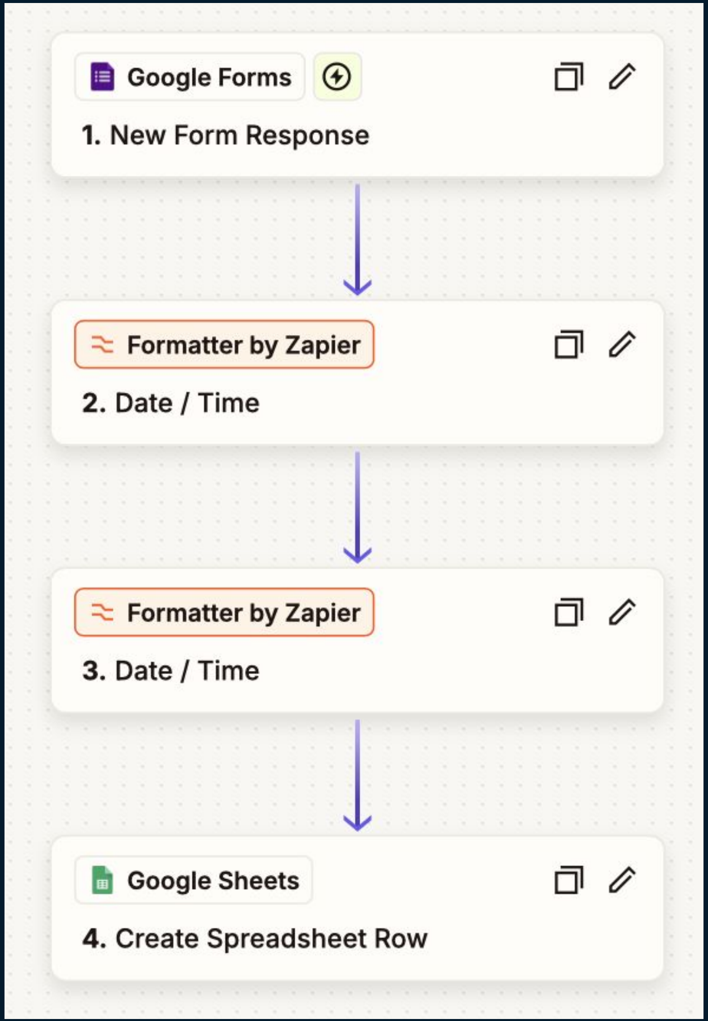
# Agentic AI Depends on Tools

- Just like us, Agents need access to tools:
    - Retrieval, summarization, communication, memory, etc.
- Tools enable capability
    - Building tools = Building agency

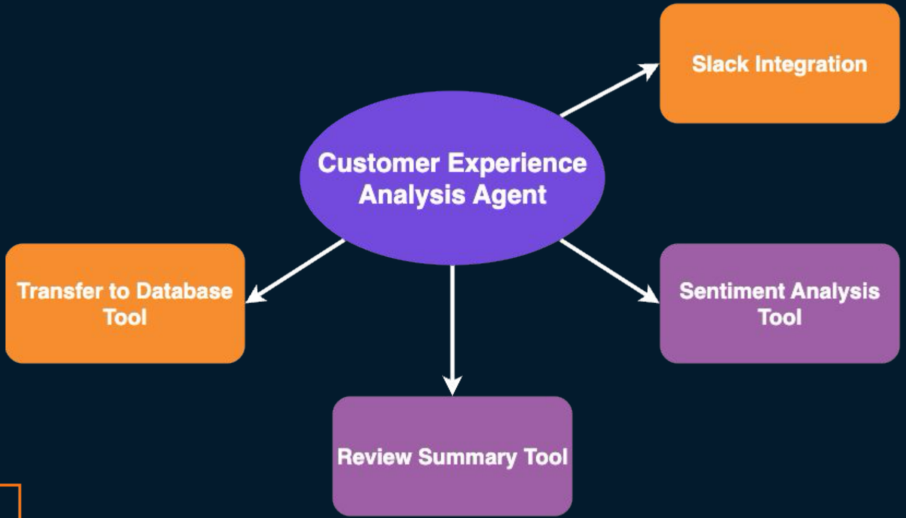


# Workflows + Agents

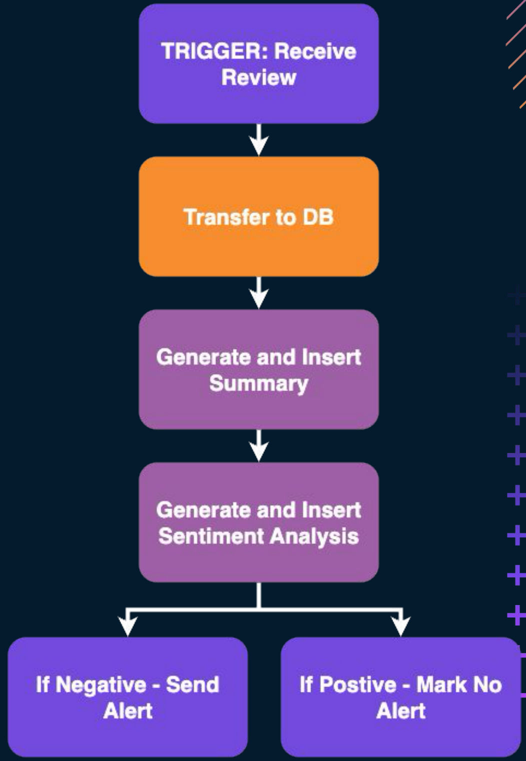
- Workflows establish execution structures
    - Trigger -> Process -> Output
    - Provides reliability, traceability, and clarity
- These form the base layer that Agents plug into
    - When LLMs join the loop, systems *think* instead of just *running*



## Agentic Tool View




## Workflow Demo



# Two Layers of Intelligence

**Reasoning Layer - Agents**

Goal-driven systems that plan and take actions

**Execution Layer - Workflows**

Defined, rule-based processes for repetition and consistency

# Zapier Demo

Applying AI Workflows

# Wrapping Up and Wednesday Preview

# What You'll Be Doing

1. Pick a **role**
2. Choose a **knowledge source**
3. **Write + Refine** prompts to simulate agentic behavior
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5. Build a system with clear inputs and outputs